

YIFAN JING

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EDUCATION

University of Michigan, B.S. in Data Science

Expected 2027

Research Interests: Multimodal Reasoning, LLM Reasoning, AI Scientist Systems, AI Evaluation

PUBLICATIONS / PREPRINTS / MANUSCRIPTS

Manuscript under review at ICLR 2027 (title withheld).

Yifan Jing, Emma Yicheng Lou, Shaotian Sun, and Yuxuan Wu. *Computing Tools for Translation-Invariant Total Orders*. arXiv preprint, 2026. [arXiv:2607.11709](https://arxiv.org/abs/2607.11709). Target: **FPSAC 2027**.

RESEARCH EXPERIENCE

Z Lab, Princeton University

Jul 2026 - Present

Research Intern | Advisor: Prof. Zhuang Liu

- Investigating generation-augmented multimodal reasoning through systematic evaluations and controlled ablations to understand when and why generated visual thoughts help or hurt VLM reasoning across models and benchmarks.
- Developing and leading a research direction on **AI scientist systems** and self-evolving agents that autonomously propose, evaluate, and iteratively improve reasoning strategies and scientific solutions.

Terminal-Bench Science

Apr 2026 - Present

Contributor

- Developing mathematical discovery and formal reasoning tasks to evaluate frontier AI agents' ability to derive nontrivial lemmas from foundational definitions.
- **1 task accepted for inclusion:** Cambrian projection in Lean 4; domain, technical, and final reviews approved as of **September 22, 2026**.
- Built a complete reference proof and isolated verifier for Reading's maximality theorem on arbitrary Coxeter systems.

[Task and review history: PR #1257](#)

SURE 2026, University of Michigan

May 2026 - Aug 2026

Undergraduate Researcher | Advisor: Prof. Lei Ying

- Developed **HintFlow**, training a 4B language-model policy to guide frozen 14B-20B reasoning models through selective supervised fine-tuning and a modified GRPO objective with virtual-positive normalization for all-failed groups.
- Built the end-to-end training and inference pipeline with Qwen3-4B, Qwen3-14B, GPT-OSS-20B, LoRA, and vLLM.

Contact Sensing Hexapod, BIRDS Lab, University of Michigan

Jan 2026 - Present

Undergraduate Researcher | Advisor: Prof. Shai Revzen

- Developed large-scale MuJoCo simulations for multi-legged locomotion, evaluating modular controllers across varying morphologies and irregular terrain conditions.
- Developing an ultra-lightweight control pipeline: train a high-capacity neural policy to near-optimal performance, then use LLM-guided search to distill it into an interpretable finite-state machine for efficient onboard control.

[Simulation contributor: arXiv:2603.09147](#)

RESEARCH EXPERIENCE (CONTINUED)

Hybrid Transformer Analysis in Induction Head Problem University of Michigan

Sep 2025 - Dec 2025

Research Project

- Studied sequence-model generalization on the Associative Recall task, comparing Transformer, Mamba, and hybrid architectures.
- Proposed and evaluated hybrid architectures combining state-space models with attention, demonstrating improved long-range generalization.
- Designed controlled experiments to analyze convergence and generalization dynamics across model families.

[Code: Analysis-AR-problem](#)

TITO (Translation-Invariant Total Orders) University of Michigan

Sep 2025 - Apr 2026

Researcher | Advisor: Dr. Grant Barkley

- Developed a new algorithmic approach for translation-invariant total orders, formalizing order-comparability through inversion sets and proving its correctness.
- Translated the theoretical model into a computable structure for efficient sequence comparison and ranking, with potential applications in sorting and discrete optimization.
- Released a SageMath/Python package for weak order comparison, join computation, normalization, and visualization; completed the research project.

[SageMath package](#) | [LoGM project](#)

SELECTED RESEARCH RESULT

General Counterexamples to a Candidate Pure-Loss Second-Order Converse

Sep 23, 2026

- With **Yue Tu** and **six parallel AI agents**, constructed a general family of counterexamples in quantum communication and verified the disproof in **Lean 4**; fixed-photon condensate codes satisfy the photon-number constraint with zero leakage.
- The candidate formulation arose from a question raised by **Mark M. Wilde, Joseph M. Renes, and Saikat Guha in 2014**, 12 years before this result. QIQOP credits Yue Tu and Yifan Jing and lists the problem as solved.

[Problem and solution](#) | [Lean 4 proof](#)

TEACHING EXPERIENCE

University of Pittsburgh, Pittsburgh, Pennsylvania Undergraduate Teaching Assistant | Intermediate Programming

Aug 2024 - Dec 2024

- Lectured an undergraduate class of 125+ students; taught Java programming in weekly office hours, requested recitations, and exam-preparation sessions.

HONORS & LEADERSHIP

Tartanhacks, Carnegie Mellon University

2026

- Led four undergraduate students and won a **Top 5 grant** among 300+ submitted projects and 700 participants.

[Project: SpaceOverflow](#)

CAST-USA 33rd Innovation Summit

- Received a **Finalist Award**, presented by Prof. Avi Wigderson, the 2023 ACM A.M. Turing Award Laureate.
- Presented the CEEG AI Agent business plan to an audience of 300.

SKILLS

Languages: Python (JAX, PyTorch, NumPy), C++, Java, HTML, JavaScript

Engineering: MuJoCo, Linux

Tools: GitHub Actions, Jupyter Notebook, Colab, Google Cloud Platform, LaTeX, Coding Agents